Home Credit Default Risk - Exploratory Data Analysis

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1. Business Problem

Home Credit provides loans to individuals who may have limited or insufficient credit histories. A key business challenge is identifying applicants who are more likely to experience difficulty repaying their loans. Better understanding the characteristics associated with repayment risk can help support more informed lending decisions and risk management.

In this project, the target variable indicates whether an applicant experienced repayment difficulties. This exploratory data analysis examines applicant characteristics, data quality, missingness, and relationships with the target. The findings will be used to identify potentially useful variables and guide feature selection and modeling decisions in the next stage of the project.

2. Data Overview

```
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```

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$ TARGET                      <dbl> 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,…
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$ AMT_ANNUITY                 <dbl> 24700.5, 35698.5, 6750.0, 29686.5, 21865.…
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$ DAYS_ID_PUBLISH             <dbl> -2120, -291, -2531, -2437, -3458, -477, -…
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$ AMT_REQ_CREDIT_BUREAU_YEAR   <dbl> 1, 0, 0, NA, 0, 1, 2, 0, 1, NA, 0, 2, 0, …
```

The training dataset contains 307,511 loan applications and 122 variables. The variables include numerical measures, categorical applicant characteristics, loan information, and the binary target variable indicating repayment difficulty. Because the dataset contains a large number of features, this EDA will focus on data quality and selected variables that may show meaningful relationships with the target.

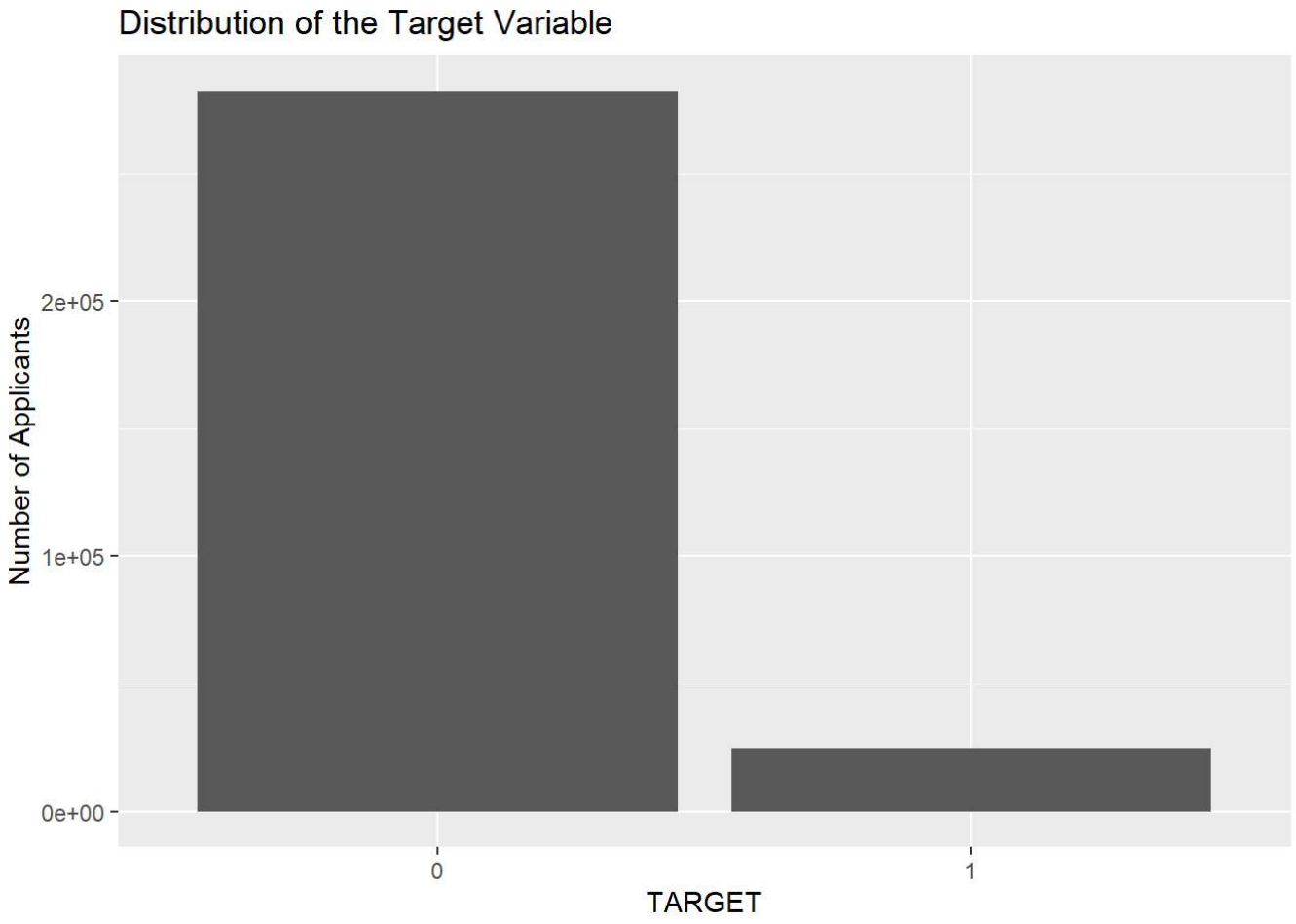
3. Target Variable

Target Distribution

```
     0      1
282686  24825


         0         1
0.91927118 0.08072882
```

TARGET = 1 represents applicants with repayment difficulties, while TARGET = 0 represents applicants without repayment difficulties. The target variable is highly imbalanced, with only about 8% of applicants in the TARGET = 1 group.



The visualization confirms the strong class imbalance in the target variable. This imbalance may need to be considered when evaluating predictive models later in the project.

4. Missingness

```
# A tibble: 10 × 2
   variable                missing_rate
   <chr>                          <dbl>
 1 COMMONAREA_AVG                 0.699
 2 COMMONAREA_MODE                0.699
 3 COMMONAREA_MEDI                0.699
 4 NONLIVINGAPARTMENTS_AVG        0.694
 5 NONLIVINGAPARTMENTS_MODE       0.694
```

```
 6 NONLIVINGAPARTMENTS_MEDI        0.694
 7 FONDKAPREMONT_MODE              0.684
 8 LIVINGAPARTMENTS_AVG            0.684
 9 LIVINGAPARTMENTS_MODE           0.684
10 LIVINGAPARTMENTS_MEDI           0.684
```

Several variables have very high missing rates, with the top variables missing approximately 68% to 70% of observations. Many of these variables are related to housing or apartment characteristics.

Missingness by TARGET

To determine whether missingness may contain useful information, I compared the missing rates of all variables across the two TARGET groups.

```
# A tibble: 10 × 4
   variable                     TARGET_0 TARGET_1 difference
   <chr>                           <dbl>    <dbl>      <dbl>
 1 EMERGENCYSTATE_MODE             0.468    0.544     0.0759
 2 TOTALAREA_MODE                  0.477    0.552     0.0755
 3 ENTRANCES_AVG                   0.497    0.572     0.0750
 4 ENTRANCES_MODE                  0.497    0.572     0.0750
 5 ENTRANCES_MEDI                  0.497    0.572     0.0750
 6 FLOORSMAX_AVG                   0.492    0.567     0.0750
 7 FLOORSMAX_MODE                  0.492    0.567     0.0750
 8 FLOORSMAX_MEDI                  0.492    0.567     0.0750
 9 YEARS_BEGINEXPLUATATION_AVG     0.482    0.556     0.0744
10 YEARS_BEGINEXPLUATATION_MODE    0.482    0.556     0.0744
```

Applicants with TARGET = 1 generally have more missing information in several housing-related variables than applicants with TARGET = 0. The difference is around 7 to 8 percentage points for the variables with the largest gaps. This suggests that missing housing information may be related to repayment difficulty and could be useful to consider in later modeling.

5. Numerical Variable Exploration

AMT_INCOME_TOTAL by TARGET

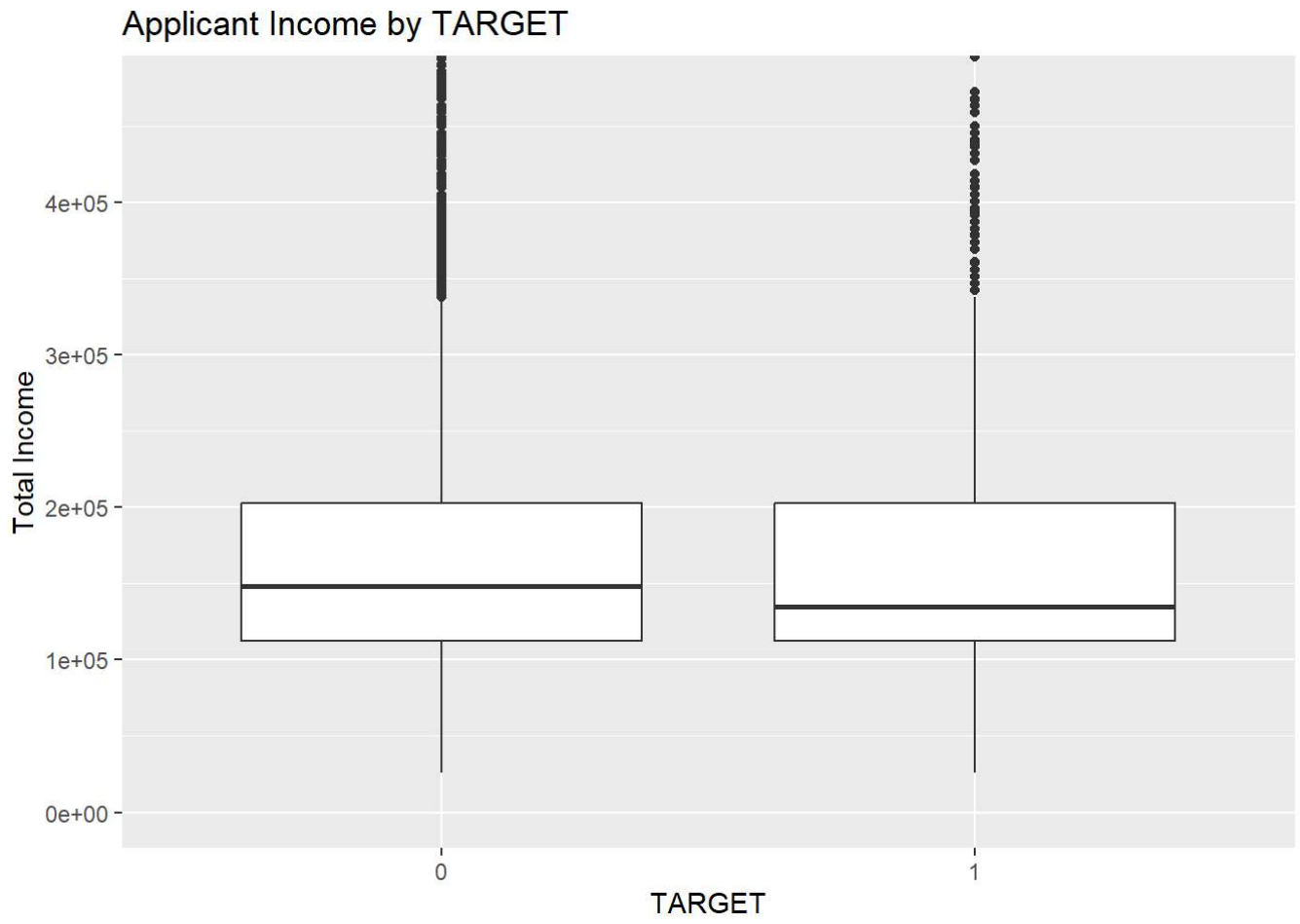
I examined whether applicant income differs between applicants with and without repayment difficulties.

```
# A tibble: 2 × 3
  TARGET mean_income median_income
   <dbl>       <dbl>         <dbl>
1      0     169078.        148500
2      1     165612.        135000
```

Applicants with TARGET = 1 have a somewhat lower median income than those with TARGET = 0. However, the difference is not very large, suggesting that income alone may not strongly distinguish applicants who experience repayment difficulties.

```
     0%      25%      50%      75%      95%      99%     100% 
  25650   112500   147150   202500   337500   472500 117000000 
```

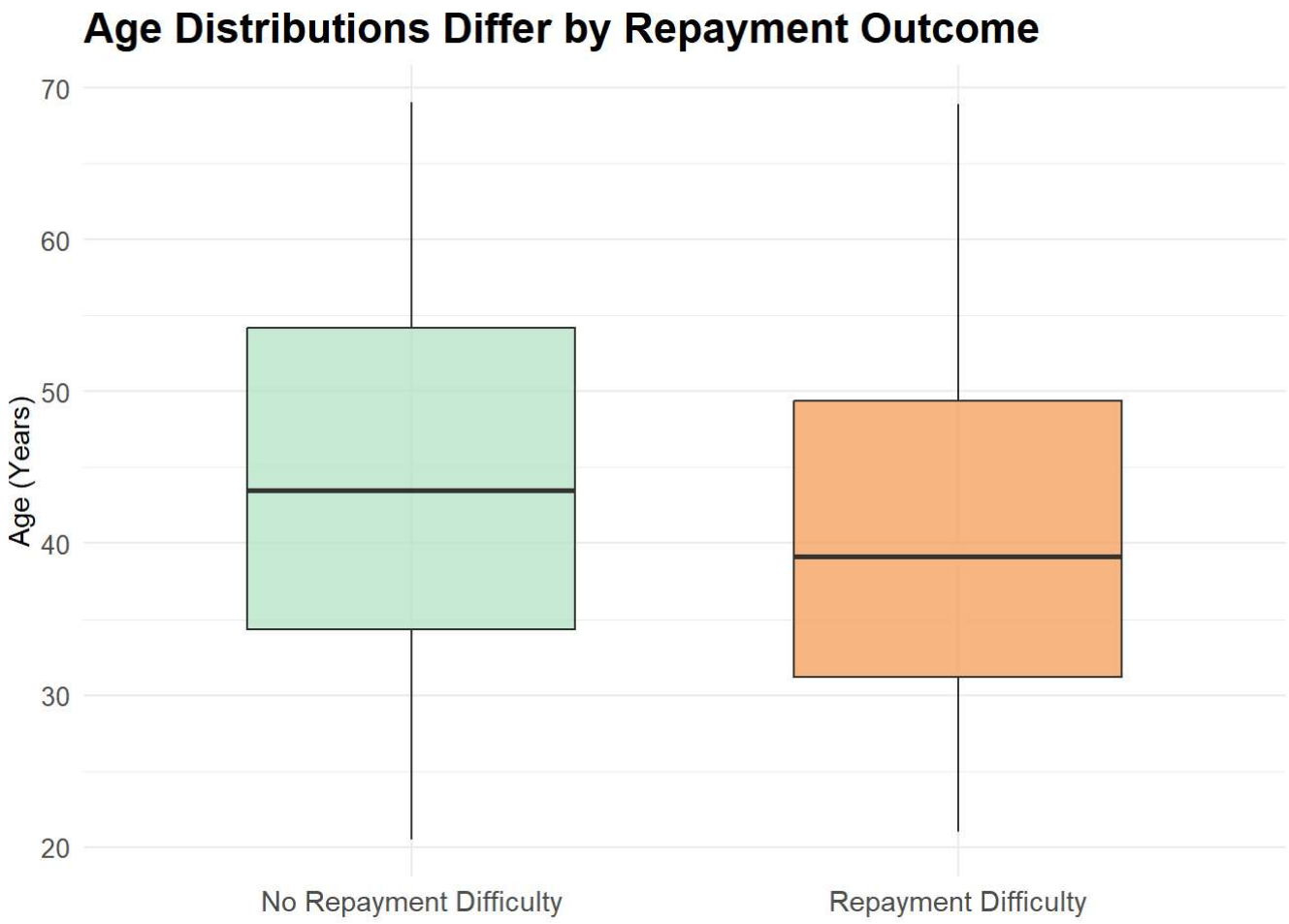
Applicant income increases gradually across the 25th to 99th percentiles, but the maximum value is dramatically higher than the 99th percentile. This suggests the presence of extreme high-income outliers, which may strongly influence the mean and distort visualizations of the income distribution.



Applicants with TARGET = 1 tend to have slightly lower income, but the difference between the two groups is not large.

Applicant Age by TARGET

```
# A tibble: 2 × 3
  TARGET mean_age median_age
   <dbl>    <dbl>      <dbl>
1      0     44.2       43.5
2      1     40.8       39.1
```



Applicants with TARGET = 1 tend to be younger than those with TARGET = 0, as shown by the lower median age and overall lower age distribution. This suggests that younger age may be associated with a higher risk of repayment difficulty.

AMT_CREDIT by TARGET

```
# A tibble: 2 × 3
  TARGET mean_credit median_credit
   <dbl>       <dbl>         <dbl>
1      0     602648.        517788
2      1     557779.        497520
```

Applicants with TARGET = 1 tend to have slightly lower credit amounts than those with TARGET = 0, The difference between the two groups is relatively small, suggesting that credit amount alone may not strongly distinguish applicants with repayment difficulties.

Credit-to-Income Ratio

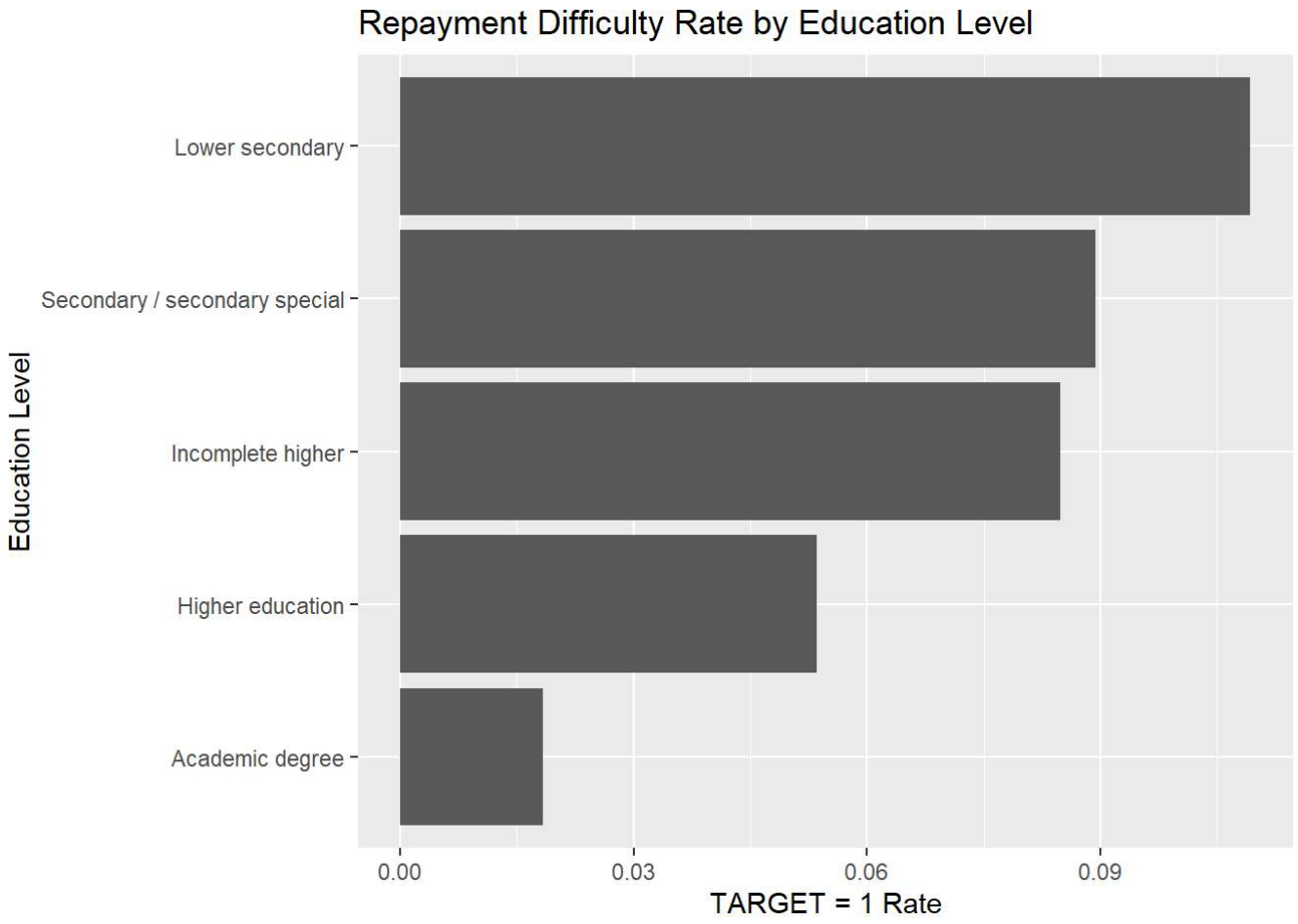
```
# A tibble: 2 × 2
  TARGET median_credit_income_ratio
   <dbl>                      <dbl>
1      0                       3.27
2      1                       3.25
```

The median credit-to-income ratio is very similar between TARGET = 0 and TARGET = 1 (3.27 vs. 3.25), suggesting that this ratio does not show a clear difference between applicants with and without repayment difficulties.

6. Categorical Variable Exploration

Education

```
# A tibble: 5 × 3
  NAME_EDUCATION_TYPE           applicants target_rate
  <chr>                              <int>       <dbl>
1 Lower secondary                     3816      0.109 
2 Secondary / secondary special     218391      0.0894
3 Incomplete higher                  10277      0.0848
4 Higher education                   74863      0.0536
5 Academic degree                      164      0.0183
```



Applicants with lower education levels tend to have higher repayment difficulty rates, suggesting that education level may be associated with repayment risk.

```
# A tibble: 5 × 4
  NAME_EDUCATION_TYPE  sample_size repayment_difficulty…¹ repayment_difficulty…²
  <chr>                      <int>                  <int>                  <dbl>
1 Lower secondary             3816                    417                 0.109 
2 Secondary / seconda…      218391                  19524                 0.0894
3 Incomplete higher          10277                    872                 0.0848
4 Higher education           74863                   4009                 0.0536
5 Academic degree              164                      3                 0.0183
# ℹ abbreviated names: ¹repayment_difficulty_count, ²repayment_difficulty_rate
```

The pattern generally shows higher repayment difficulty rates among lower education levels, although the Academic degree group has a very small sample size and should be interpreted cautiously.

Income Type

```
# A tibble: 8 × 3
  NAME_INCOME_TYPE     applicants target_rate
  <chr>                     <int>       <dbl>
1 Maternity leave               5      0.4
2 Unemployed                   22      0.364
3 Working                  158774      0.0959
4 Commercial associate      71617      0.0748
5 State servant             21703      0.0575
6 Pensioner                 55362      0.0539
7 Businessman                  10      0
8 Student                      18      0
```

Among the major income groups, working applicants have the highest repayment difficulty rate, followed by commercial associates, while state servants and pensioners have lower rates. Very high or zero rates in small groups should be interpreted cautiously because of their limited sample sizes.

Housing Type

```
# A tibble: 6 × 3
  NAME_HOUSING_TYPE   applicants target_rate
  <chr>                    <int>       <dbl>
1 Rented apartment          4881      0.123
2 With parents             14840      0.117
3 Municipal apartment      11183      0.0854
4 Co-op apartment           1122      0.0793
5 House / apartment       272868      0.0780
6 Office apartment          2617      0.0657
```

Applicants living in rented apartments or with parents have higher repayment difficulty rates than those living in a house or apartment. This suggests that housing type may be associated with repayment risk. However, this is a descriptive relationship and should not be interpreted as causal.

7. Results and Key Findings

Data Characteristics and Quality Issues

The EDA identified several important data quality issues. First, the TARGET variable is highly imbalanced, with only about 8% of applicants experiencing repayment difficulties. Several housing-related variables also contain substantial missingness, with some variables missing nearly 70% of observations. In addition, applicant income contains extreme high-value outliers, which can strongly influence the mean and distort visualizations.

Missingness itself may also contain useful information. For several housing-related variables, applicants with TARGET = 1 had missing rates approximately 7 to 8 percentage points higher than applicants with TARGET = 0. This suggests that whether housing information is missing may itself be associated with repayment difficulty.

Key Relationships with TARGET

Several variables showed meaningful patterns with repayment difficulty. Applicants with TARGET = 1 tended to be younger than applicants with TARGET = 0, suggesting that age may be an informative predictor. Education level also showed differences in repayment difficulty rates, with lower education levels generally having higher rates. Housing type and income type also showed differences across applicant groups, although categories with small sample sizes should be interpreted cautiously.

Other variables showed weaker relationships. Applicant income was somewhat lower among TARGET = 1 applicants, but the difference between the two groups was relatively small. Credit amount was also only slightly different between the groups. The credit-to-income ratio was nearly the same for TARGET = 0 and TARGET = 1, suggesting that this engineered feature may not provide strong separation on its own.

Implications for Modeling

These findings suggest several considerations for the next stage of the project. Age, education level, income type, and housing type may be useful features to include in predictive models. Because missingness in some housing variables differs by TARGET, creating missing-value indicators may also be useful rather than treating missing values only as a data-cleaning problem.

The extreme income values may require transformation or other robust treatment before modeling. The strong class imbalance in TARGET should also be considered when evaluating model performance, because accuracy alone may provide a misleading picture of model quality. Finally, variables that showed only small differences between TARGET groups, such as credit amount and the credit-to-income ratio, may still contribute when combined with other variables but appear to have limited predictive value when considered individually.

8. Use of AI

I used the instructor’s slides and assignment guidelines as the main framework for this EDA. The slides provided suggestions about what to examine, such as the target variable, missingness, numerical and categorical variables, and relationships with the target. I then used ChatGPT to help turn these suggestions into a more detailed EDA plan for the Home Credit dataset.

During the analysis, ChatGPT helped me understand R functions, generate and troubleshoot code, and interpret the outputs. I discussed the results with AI step by step and used the actual data results to decide which relationships were meaningful and which were relatively weak.

Overall, AI helped me apply the ideas from the course materials more efficiently and helped me better understand the reasoning behind each step of the EDA.